

吴筱琦

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现居城市：伦敦/苏州

意向城市：上海/苏州/杭州 | 数据分析师/数据科学家



教育背景

帝国理工学院 (QS 2) 计算生命科学研究型硕士

伦敦 2025.09 - 2026.09

帝国理工学院 (QS 2) 生物科学本科

伦敦 2022.09 - 2025.07

• 相关课程：编程与统计，生物信息学，遗传学，细胞生物学与遗传学，应用分子生物学

实习经历

上海宣泰医药科技有限公司 分析工程师 · 分析部

上海 2024.07 - 2024.09

- **数据分析**：独立完成客户样本的前处理、HPLC 系统操作及数据分析，在严格的质控流程下确保检测结果的准确性和可靠性
- **流程执行**：使用 HPLC/UV 技术对药品样本进行稳定性检测，系统鉴别潜在杂质，为分析结论提供可追溯的数据支持
- **质量控制**：通过严格的质控流程系统识别潜在杂质，为产品质量决策提供关键数据支持，展现了保证数据准确性与可追溯性的工程化思维

科研与项目经历

多模态大语言模型提取媒介性状数据评估：高精确率与结构依赖型记录召回

伦敦 2025.12 - 2026.08

Python、Gemma 4 31B IT (多模态 LLM)、PyMuPDF、Docling、提示工程、JSON Schema 校验、pytest

- **基准构建与提取流程**：设计加权多样性贪心算法从 VecTraits 超 4 万行、334 篇论文中选取 20 篇，构建 2,335 条开发基准与 431 条留出测试集；结合 PyMuPDF/Docling 证据提取，驱动 Gemma 4 31B IT 完成提示工程约束下的性状记录提取，经 11 轮迭代取得开发集条件宏观字段 F1 0.810
- **严谨评估**：设计二分图记录匹配评估框架（行集合/条件字段/端到端字段），在留出测试集上验证锁定流程，取得记录精确率 1.000、召回率 0.360、条件宏观字段 F1 0.729

气候变化对佛罗里达州埃及伊蚊种群动态的影响建模

伦敦 2025.03 - 2025.06

R、deSolve、ggplot2、ODEs、统计验证 (RMSE、Pearson、Spearman)

- **模型适配**：将已发表的热带 ODE 房室模型适配至亚热带地理环境，模拟佛罗里达 Collier 县埃及伊蚊种群在温度与降雨影响下的动态变化
- **模型验证**：基于 53 周本地蚊虫监测数据验证模型 (RMSE=1.65, Pearson $r=0.29$, Spearman $\rho=0.30$)，优化时间滞后 ($\tau = -2$ 周) 与降雨尺度因子，量化模型性能
- **研究成果**：提出模型改进方向（引入人为水源、空间异质性与不确定性量化），展示端到端建模与数据验证能力

生态建模与种群动力学模拟

伦敦 2024.01 - 2024.02

R、ggplot2、Lotka-Volterra、逻辑斯蒂增长模型

- **建模项目组**：构建四个种群动力学模型——集合种群、渔业捕捞 (MSY)、资源竞争及三物种捕食-被捕食模型，均使用差分方程与随机扩展实现；四项均获 90+ 评分（最高 94 分），展现扎实的算法实现与参数优化能力

技能/其他

- **Python 与开发工具**：Python (主力语言)，Git，虚拟环境，Claude Code / Cursor 辅助开发；多后端 LLM 集成、提示工程、JSON Schema 校验、pytest 测试、基于 JSON 配置/状态跟踪的异步子进程编排
- **Web 开发**：FastAPI (路由/服务架构)、React、TypeScript、Vite、Tailwind CSS
- **数据分析与建模**：R (ggplot2、dplyr、caper、deSolve)、统计推断 (GLM、PGLS、Wilcoxon 检验、t 检验、Pearson/Spearman 相关)、基于微分方程的动态系统建模与参数优化
- **语言**：英语 (雅思 7.5)，普通话 (母语)，法语 (A2)；**兴趣爱好**：钢琴，滑雪

Xiaoqi Wu

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Currently based in: London / Suzhou

Data Analyst | Data Scientist | Shanghai / Suzhou / Hangzhou

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Programming and Statistics, Bioinformatics, Genetics, Cell Biology and Genetics, Applied Molecular Biology

Internship Experience

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Engineer · Analytical Department Shanghai 2024.07 - 2024.09

- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis on client-supplied samples, ensuring accuracy and reliability under strict quality-control procedures
- **Process Execution:** Conducted stability testing of drug samples using HPLC/UV, systematically identifying potential impurities and providing traceable data to support analytical conclusions
- **Quality Control:** Systematically identified potential impurities through rigorous quality-control procedures, providing key data to support product quality decisions – reflecting an engineering mindset toward data accuracy and traceability

Research & Project Experience

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery London 2025.12 - 2026.08

Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, bipartite record-matching evaluation, pytest

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers (spanning trait types, reporting formats, and data sources) from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set; reduced the 91-column VecTraits schema to a 22-field extraction schema aligned with the MIREVTD reporting standard
- **Multimodal Extraction Pipeline:** Built a multimodal evidence pipeline combining PyMuPDF full-text extraction, Docling table-structure parsing, and 150 DPI figure/table rendering, fed into Gemma 4 31B IT (a 256K-context multimodal LLM) to extract provenance-linked trait records from PDF text, tables, and images via prompt engineering and JSON Schema-constrained output; iterated across 11 prompt versions, with the final version reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Results:** Designed a bipartite record-matching evaluation framework (row-set, conditional-field, and end-to-end field metrics) with systematic error analysis; evaluated the locked pipeline on the held-out test set, achieving row precision of 1.000, row recall of 0.360, and conditional macro field F1 of 0.729 – confirming a "high-precision, structure-dependent recall" profile and supporting the pipeline's use as a candidate-record generator within a human-in-the-loop curation workflow

Modelling the Impact of Climate Change on *Aedes aegypti* Population Dynamics in Florida London 2025.03 - 2025.06

R, deSolve, ggplot2, ODEs, statistical validation (RMSE, Pearson, Spearman)

- **Model Adaptation:** Adapted a published tropical ODE compartment model to a subtropical geography to simulate *Aedes aegypti* population dynamics in Collier County, Florida under temperature and rainfall effects
- **Validation:** Validated the model against 53 weeks of local mosquito-surveillance data (RMSE=1.65, Pearson $r=0.29$, Spearman $\rho=0.30$), optimizing time-lag ($\tau = -2$ weeks) and a rainfall scaling factor

to quantify model performance

- **Outcome:** Proposed model improvements (anthropogenic water sources, spatial heterogeneity, uncertainty quantification), demonstrating end-to-end modelling and data-validation ability

Ecological Modelling and Population Dynamics Simulation

London 2024.01 - 2024.02

R, ggplot2, Lotka-Volterra, logistic growth model

- **Modelling Suite:** Built four population-dynamics models – metapopulation, fisheries harvesting (MSY), resource competition, and a 3-species predator-prey system, each implemented with difference equations and stochastic extensions; all four scored 90+ (up to 94), demonstrating solid algorithm implementation and parameter-optimization skills

Skills & Other

- **Python & Tooling:** Python (primary language), Git, virtual environments, Claude Code / Cursor for AI-assisted development; multi-backend LLM integration, prompt engineering, JSON Schema validation, pytest testing, async subprocess orchestration with JSON-based config/state tracking
- **Web Development:** FastAPI (routers/services architecture), React, TypeScript, Vite, Tailwind CSS
- **Data Analysis & Modelling:** R (ggplot2, dplyr, caper, deSolve), statistical inference (GLM, PGLS, Wilcoxon signed-rank, t-tests, Pearson/Spearman correlation), ODE-based dynamic system modelling and parameter optimization
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, skiing